Ford
The Universal Car

FORD MANUAL

For Owners and Operators
of Ford Cars and Trucks

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Foreword

IT is a significant fact that nearly all Ford cars are driven by laymen—by owners, who in the great majority of cases have little or no practical experience with things mechanical.

The simplicity of the Ford car and the ease with which it is operated renders an intimate knowledge of mechanical technicalities unnecessary for its operation.

And the further fact that there are more than twenty thousand Ford service stations distributed throughout the civilized world—where adjustments and repairs may be had with no annoying delay—gives to Ford owners a singular freedom from mechanical annoyances which beset owners of cars having limited service facilities and distribution.

But while it is not imperative, it is, however, altogether desirable that every Ford owner should thoroughly understand his car. With such knowledge at his command he is always master of the situation—he will maintain his car more economically—prolong its usefulness—and he will also derive more pleasure from it, for it is a truism that the more one knows about a thing the more one enjoys it.

The mastery of a thorough knowledge of Ford construction is by no means a difficult or time-consuming task. The Ford is the simplest car made. It is easy to understand, and is not difficult to keep in proper adjustment and repair.

That the Ford construction may be thoroughly understood—and that there may be an authoritative guide for the making of Ford adjustments—this book is published.

Important

It is most important that owners of Ford cars shall insist upon getting the genuine Ford-made materials, or "Parts," when having repairs or replacements made. This will be assured when the car is taken to the authorized Ford agent. Imitation, or "bogus" or counterfeit parts of inferior quality are being made and sold as "Ford Parts." Avoid their use by dealing with the authorized Ford agent.

The Car and Its Operation

What must be done before starting the Car?

Answer No. 1

Before trying to start the car, fill the radiator (by removing cap at top) with clean fresh water. If perfectly clean water cannot be obtained it is advisable to strain it through muslin or other similar material to prevent foreign matter from getting in and obstructing the small tubes of the radiator. The system will hold approximately three gallons. It is important that the car should not be run under its own power unless the water circulating system has been filled. Pour in the water until you are sure that both radiator and cylinder water jackets are full. The water will run out of the overflow pipe onto the ground when the entire water system has been properly filled. During the first few days that a new car is being driven it is a good plan to examine the radiator frequently and see that it is kept properly filled. The water supply should be replenished as often as may be found necessary. Soft rain water, when it is to be had in a clean state, is superior to hard water, which may contain alkalies and other salts which tend to deposit sediment and clog the radiator. (See chapter on Cooling System.)

What about Gasoline?

Answer No. 2

The ten-gallon gasoline tank should be filled—nearly full—and the supply should never be allowed to get low. When filling the tank be sure that there are no naked flames within several feet, as the vapor is extremely volatile and travels rapidly. Always be careful about lighting matches near where gasoline has been spilled, as the air within a radius of several feet is permeated with the highly explosive vapor. The small vent hole in the gasoline tank cap should not be allowed to get plugged up, as this would prevent proper flow of the gasoline to the carburetor. The gasoline tank may be drained by opening the pet cock in the sediment bulb at the bottom.

How about the Oiling System?

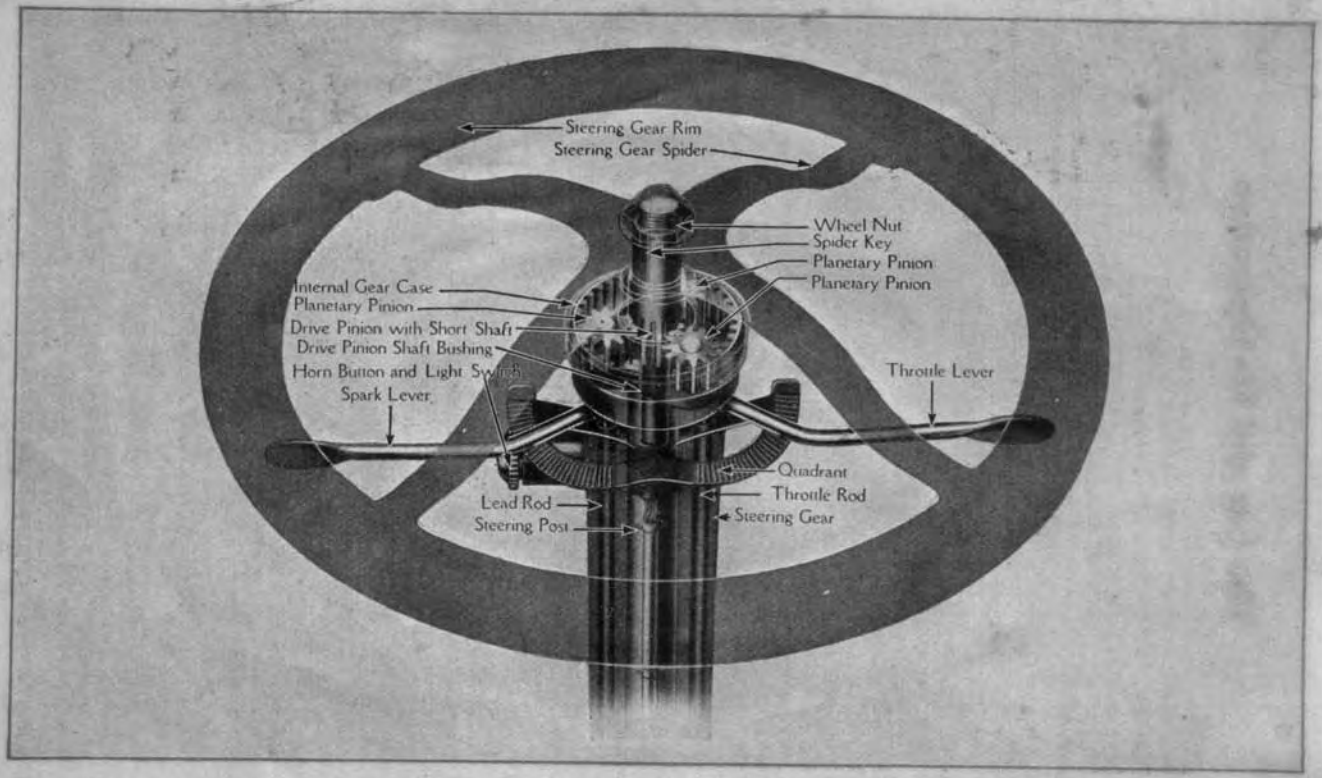
Answer No. 3

Upon receipt of the car see that a supply of medium light, high-grade gas engine oil is poured into the crank case through the breather pipe at the front of the engine (a metal cap covers it). Down under the car in the flywheel casing (the reservoir which holds this oil) you will find two pet cocks. Pour oil in slowly until it runs out of the upper cock. Leave the cock open until it stops running—then close it. After the engine has become thoroughly limbered up, the best results will be obtained by carrying the oil at a level midway between the two cocks—but under no circumstances should it be allowed to get below the lower cock. All other parts of the car are properly oiled when it leaves the factory. However, it will be well to see that all grease cups are filled and that oil is supplied to necessary parts. (See Cut No. 18, also chapter on Lubrication.)

How are Spark and Throttle Levers used?

Answer No. 4

Under the steering wheel are two small levers. The right-hand (throttle) lever controls the amount of mixture (gasoline and air) which goes into the engine. When the engine is in operation, the farther this lever is moved downward toward the driver (referred to as "opening the throttle") the faster the engine runs and the greater the power furnished. The left-hand lever controls the spark, which explodes the gas in the cylinders of the engine. The advancing of this lever



Steering Wheel, showing reduction gears meshing with the teeth of the gear case and center pinion. (Cut No. 1)

"advances the spark," and it should be moved down notch by notch until the motor seems to reach its maximum speed. If the lever is advanced beyond this point a dull knock will be noticed in the engine. (See chapter on Ignition.)

Where should these levers be when starting the Engine?

Answer No. 5

The spark lever should usually be put in about the third or fourth notch of the quadrant (the notched half-circle on which the levers operate). The throttle should usually be opened about five or six notches. A little experience will soon teach you where these levers should be placed for proper starting. Care should be taken not to advance the spark lever too far, as the engine may "back kick."

What else is necessary before starting the Engine?

Answer No. 6

First, see that the hand lever, which comes up through the floor of the car at the left of the driver, is pulled back as far as it will go. The lever in this position holds the clutch in neutral and engages the hub brake, thus preventing the car moving forward when the engine is started. Second, after inserting the switch key in the switch on the coil box, throw the switch lever as far to the left as it will go—to the point marked "magneto." This switch connects the magneto with the engine. The engine can't be started until it is on—and the throwing off of this switch stops the engine.

How is the Engine started?

Answer No. 7

If the car is not equipped with a starter the engine is started by the lifting of the starting crank at the front of the car. Take hold of the handle and push firmly toward the car till you feel the crank ratchet engage, then lift upward with a quick swing. With a little experience this operation will become an easy matter. Don't, as a usual thing, crank downward against the compression—for then an early explosion may drive the handle vigorously backward. This does not mean however, that it is not advisable, when the car is hard to start, to occasionally "spin" the engine by the use of the starting handle—but be sure the spark lever is retarded when spinning or cranking the engine against compression, otherwise a sudden backfire may injure the arm of the operator. When the engine is cool it is advisable to prime the carburetor by pulling on the small wire at the lower left corner of the radiator while giving the engine two or three quarter turns with the starting handle.

If the car is equipped with a starter the spark and throttle levers should be placed in the same position on the quadrant as when cranking by hand, and the ignition switch turned on. Current from either battery or magneto may be used for ignition. However, we recommend that the magneto be used at all times. The magneto was designed to furnish ignition for the Model T engine and better results will be obtained by operating in this way. Special attention must be paid to the position of the spark lever as a too advanced spark will cause serious backfiring which in turn will bend or break the shaft in the starter. The starting motor is operated by a push button, conveniently located in the floor of the car at the driver's feet. With the spark and throttle levers in the proper position, and the ignition switch turned on, press on the push button with the foot. This closes the circuit between the battery and starting motor, causing the pinion of the Bendix drive shaft to engage with the teeth on the flywheel, thus turning over the crankshaft. When the engine is cold it may be necessary to prime it by pulling out the carburetor priming rod, which is located on the instrument board. In order to avoid flooding the engine with an over rich mixture of gas, the priming rod should only be held out for a few seconds at a time.

How is the Engine best started in cold weather?

Answer No. 8

As gasoline does not vaporize readily in cold weather it is naturally more difficult to start the motor under such conditions. The usual method of starting the engine when cold is to turn the carburetor dash adjustment one-quarter turn to the left in order to allow a richer mixture of gasoline to be drawn into the cylinders; then hold out the priming rod, while you turn crank from six to eight one-quarter turns in quick succession, or turn the motor over a few times with the starter. Another method of starting a troublesome cold engine is as follows: Before you throw on the magneto switch, (1) close throttle lever; (2) hold out priming rod while you give crank several quick turns, or turn the motor over a few times with the starter, then let go of priming rod (being careful that it goes back all the way); (3) place spark lever in about third notch and advance throttle lever several notches; (4) throw on switch (being sure to get it on side marked "Magneto"); (5) give crank one or two turns, or close the starting switch, and the motor should start. After starting the motor it is advisable to advance the spark eight or ten notches on the quadrant and let the motor run until thoroughly heated up. If you start out with a cold motor you will not have much power and are liable to "stall." The advantage of turning on the switch last, or after priming, is that when you throw on the switch and start the motor, you have plenty of gas in the cylinders, to keep the motor running, thereby eliminating the trouble of the motor starting and stopping. After motor is warmed up turn carburetor adjustment back one-quarter turn.

To facilitate starting many drivers make a practice of stopping their engine by pulling out on the priming rod, which has the effect of shutting off the air suction and filling the cylinders full of a very rich gasoline vapor. This should not be done unless the car is going to stand over night or long enough to cool off. If the motor is stopped in this way and then started when hot, starting is apt to be difficult on account of the surplus gasoline in the carburetor.

What function does the Hand Lever perform?

Answer No. 9

Its chief purpose is to hold the clutch in neutral position. If it were not for this lever the driver would have to stop the engine whenever he left the driver's seat. He would also be unable to crank the engine without the car starting forward with the first explosion. When pulled back as far as it will go, the hand lever acts as an emergency brake on the rear wheels, by expanding the brake shoes in the rear wheel drums. Therefore the hand lever should be back as far as it will go when cranking the engine or when the car is at rest. It should be only in a vertical position, and not far enough backward to act as a brake on the rear wheels, when the car is to be reversed. When the car is operating in high or low speed the hand lever should be all the way forward.

How do the Foot Pedals operate?

Answer No. 10

The first one toward the left operates the clutch. When pressed forward the clutch pedal engages the low speed. When half-way forward the clutch is in neutral (i. e., disconnected from the driving mechanism of the rear wheels), and the releasing of this pedal engages the high-speed clutch. The center pedal operates the reverse. The right-hand pedal operates the transmission brake.

How is the Car started?

Answer No. 11

Slightly accelerate the engine by opening the throttle, press the clutch pedal half way forward, thereby holding the clutch in a neutral position while

throwing the hand lever forward; then press the pedal forward into slow speed and when under sufficient headway (20 to 30 feet), allow the pedal to drop back slowly into high speed, at the same time partially closing the throttle, which will allow the engine to pick up its load easily. With a little practice, the change of speeds will be easily accomplished, and without any appreciable effect on the smooth running of the machine.

How is the Car stopped?

Answer No. 12

Partially close the throttle; release the high speed by pressing the clutch pedal forward into neutral; apply the foot brake slowly but firmly until the car comes to a dead stop. Do not remove foot from the clutch pedal without first pulling the hand lever back to neutral position, or the engine will stall. To stop the motor, open the throttle a trifle to accelerate the motor and then throw off the switch. The engine will then stop with the cylinders full of explosive gas, which will naturally facilitate starting.

Endeavor to so familiarize yourself with the operation of the car that to disengage the clutch and apply the brake becomes practically automatic—the natural thing to do in case of emergency

How is the Car reversed?

Answer No. 13

It must be brought to a dead stop. With the engine running, disengage the clutch with the hand lever and press the reverse pedal forward with the left foot, the right foot being free to use on the brake pedal if needed. Do not bring the hand lever back too far or you will set the brakes on rear wheels. Experienced drivers ordinarily reverse the car by simply holding the clutch pedal in neutral with the left foot, and operating the reverse pedal with the right.

How is the Spark controlled?

Answer No. 14

By the left-hand lever under the steering wheel. Good operators drive with the spark lever advanced just as far as the engine will permit. However, advancing the spark too far will cause a dull knock in the motor, due to the fact that the explosion occurs too early. The spark should only be retarded when the engine slows down on a heavy road or steep grade, but care should be exercised not to retard the spark too far as this will result in late ignition, which causes loss of power and overheating of the motor and may also result in warped, burned or cracked valves. Learn to operate the spark as the occasion demands. The greatest economy in gasoline consumption is obtained by driving with the spark advanced sufficiently to obtain the maximum speed.

How is speed of Car controlled?

Answer No. 15

The different speeds required to meet road conditions are obtained by opening or closing the throttle. Practically all the running speeds needed for ordinary travel are obtained on high gear, and it is seldom necessary to use the low gear except to give the car momentum in starting. The speed of the car may be temporarily slackened in driving through crowded traffic, turning corners, etc., by "slipping the clutch," i. e., pressing the clutch pedal forward into neutral.

Is it advisable for owners to make their own Adjustments?

Answer No. 16

The Ford is the simplest of all cars. Most of the ordinary adjustments an owner will soon learn to make for himself. But we must strongly recommend that when it becomes necessary to employ the services of a mechanic, the car be taken to a Ford mechanic—one of our own representatives who thoroughly understands the car—and who will have no motive for running up useless repair bills. The entire Ford organization is interested in keeping every individual Ford car in constant operation, at the lowest possible cost. We have known of much damage done to many cars by unskilled repair men.

What attention does the Car need?

Answer No. 17

Remember that a new machine requires more careful attention during the first few days it is being driven than after the parts have become thoroughly "worked in." The car which is driven slowly and carefully when new usually gives the most satisfactory service in the end. Never start out with your car until you are sure that is has plenty of oil and water. Frequently inspect the running gear. See that no unnecessary play exists in either front or rear wheels, and that all bolts and nuts are tight. Make a practice of taking care of every repair or adjustment as soon as its necessity is discovered. This attention requires but little time and may avoid delay or possible accident on the road. We aim to deliver the car in proper mechanical adjustment. Afterwards it is plainly the duty of the driver to keep it in that condition.

The Ford Engine

What is the principle of the gasoline-driven Engine?

Answer No. 18

Gasoline when mixed with air and compressed is highly explosive. An explosion is a violent expansion caused by instantaneous combustion of confined gases. In the gasoline engine the mixture is drawn into a cylinder, where it is compressed by an advancing piston and then exploded by an electric spark, which sends the piston downward, and through the connecting rod imparts a rotary motion to the crankshaft. (See Cut No. 2.)

What are functions of the Pistons?

Answer No. 19

On the downward stroke the suction of the piston draws the fresh gas from the carburetor, through the inlet pipe and valve, into the cylinder. The upward movement of the piston compresses the gas into a very small space, between the top of the piston and the depression in the cylinder head, known as the "combustion chamber." (The compressed gases exert a pressure of approximately 60 pounds per square inch.) At this point the electric spark, generated by the magneto, explodes the gases—driving the piston downward—thus producing the power which turns the crank shaft. On the next stroke upward the piston drives the exploded gas out through the exhaust valve and pipe to the muffler. (See Cut No. 3.)

Connecting Rod—how removed?

Answer No. 20

It is a steel rod, connecting piston and crank shaft. Should the babbitt bearing become worn, or burned out through lack of oil, a knocking in the engine will result—in which case the entire connecting rod should be replaced. To make this replacement, (1) drain oil from crank case; (2) take off cylinder head; (3) remove detachable plate on bottom of crank case; (4) disconnect connecting rod from crank shaft; (5) take piston and rod out through top of cylinder.

What is the Valve Arrangement?

Answer No. 21

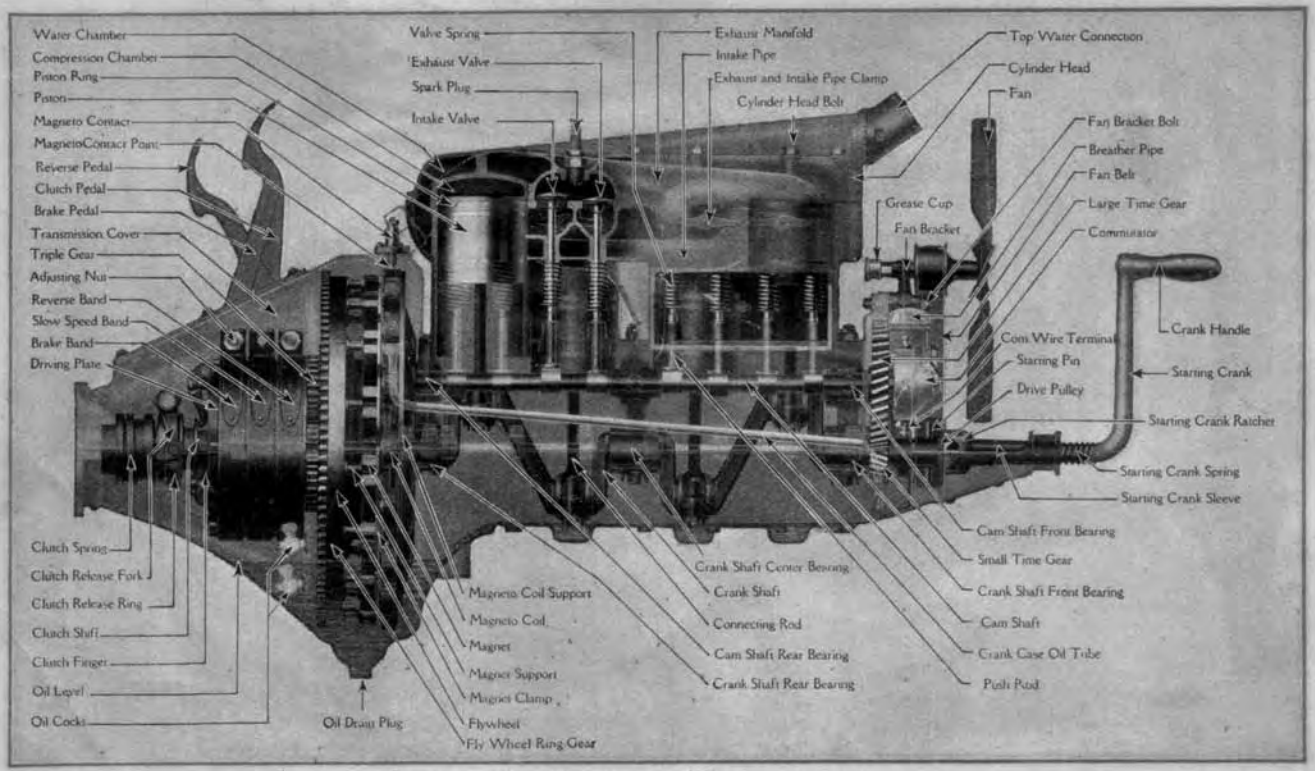
One intake and one exhaust valve are located in each cylinder. The former admits the fresh gas drawn from the carburetor through the inlet pipe—the latter permits the exploded gas to be driven out through the exhaust pipe. The valves are alternately opened and closed (see Cut No. 3) by the cams on the cam shaft striking against push rods which in turn lift the valves from their seats.

What about Valve timing?

Answer No. 22

In timing the engine the points of opening and closing of the valves are, of course, what should be considered. As the valves are properly timed at the factory when the engine is built the necessity for retiming would occur only when such parts as the cam shaft, time gears or valves were removed in overhauling the engine.

In fitting the large time gear to the cam shaft it is important to see that the first cam points in a direction opposite to the zero mark (see Cut No. 3). The time gears must also mesh so that the tooth marked zero (0) on the small time gear will come between the two teeth on the large gear at the zero point. The time gears



Sectional View of the Model T Motor. (Cut No. 2)